INCH-POUND
MIL-DTL-24640/21D
22 August 2002
SUPERSEDING
MIL-C-24640/21C
26 June 1995

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, FOUR-CONDUCTOR, LOW POWER, TYPE FXW (INCLUDING VARIATION TYPE FXOW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24640 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

Construction (Watertight)

First - Copper conductor, uncoated in accordance with ASTM B 8, 7 strands (watertight). AWG size and dimensions in accordance with Tables I and II for the applicable type.

Second - Mica glass tape insulation 0.0029 ± .0002 inch (50 percent overlap maximum and minimum overlap of 45 percent). Quality conformance testing shall be in accordance with MIL-W-81044, prior to cabling.

Third - Opaque-pigmented polyimide tape .001 inch (50 percent overlap minimum).

Fourth - Opaque-pigmented polyimide.001 inch with 0.0001 inch Fluorinated Ethylene Propylene [FEP] tape (50 percent overlap minimum).

Fifth - Modified aromatic polyimide resin coating 0.0005 inch minimum (.001 inch nominal). Identification codes applied using Method 1 or 3.

CONSTRUCTION VARIANT FOR TYPE FXW

Sixth - Four component wires shall be cabled together. Fillers may be used as necessary, to form a round cable.

Seventh - Binder tape applied helically with overlap.

Eighth - Cross-linked polyolefin jacket. (See Table I for thickness).

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CONSTRUCTION VARIANT FOR TYPE FXOW

Eighth - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.
Ninth - Binder tape applied helically with overlap.
Tenth - Cross-linked polyolefin jacket. (See Table II for thickness).

TABLE I. Construction details.

PIN M24640/20	Type number	AWG size	Conductor diameter (inch) (nom)	Component wire diameter (inch) (min)	(max)	Jacket thickness (inch) (min)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)	Conductor resistance per 1000 feet (ohms)
-01UN -02UN	FXW-3 FXW-4	16 14	0.058 0.073	0.075 0.091	0.081 0.097	0.034 0.038	0.266 0.315	0.286 0.339	78.2 115.1	4.35 2.88

TABLE II. Construction details - overall shield.

PIN M24640/20	Type number	AWG size	Conductor diameter (inch) (nom)	Component wire diameter (inch) (min)	(max)	Diameter over shield (inch) (nom)	Jacket thickness (inch) (min)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)	Conductor resistance per 1000 feet (ohms)
-01UO	FXOW-3	16	0.058	0.075	0.081	0.240	0.038	0.324	0.350	131.6	4.35
-02UO	FXOW-4	14	0.073	0.091	0.097	0.262	0.041	0.366	0.394	164.9	2.88

REQUIREMENTS:

Qualification required.
Construction details: See Tables I and II.

Component wire: For polyimide tape constructions, the following additional tests shall be applicable in accordance with MIL-W-81381:
Temperature rating: 150°C (302°F) maximum conductor temperature.
Voltage rating: 600 volts (rms) at sea level.
Blocking: Oven temperature, 200 ± 2°C (392 ± 3.6°F) no sticking of conductors.
Flammability: 3 seconds (max) after flame.
3.0 inches (max) flame travel.
No flaming of tissue paper.
Humidity resistance: 5 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test: 100 percent test. Impulse voltage at 8.0 kV.
Insulation resistance: 2500 megohms for 1000 feet (min).

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Polyimide cure test: No cracking.
Shrinkage: 0.062 inch maximum at 180°C.
Surface resistance: 5 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
Maximum change in measurement: 0.031 inch.
Wet dielectric test: 2500 volts (rms).

EXAMINATION AND TESTS:

Basic electrical:	Requirements:
Conductor Resistance - ohms/1000 feet at 25°C, maximum	See Tables I and II
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield (FXOW only)	1000
Shield to shield	Not applicable
Insulation resistance - megohms-1000 feet, minimum	
Conductor to conductor	2500
Conductor to overall shield (type FXOW only)	2500
Shield to shield (component shield only)	Not applicable
Shield to shield (component shield to overall shield)	Not applicable
Conductor and shield continuity	Not applicable
Jacket flaws (type FXOW only)	No failure

Group A:

Visual and dimensional	No failure

Capacitance:

Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	Not applicable
Capacitance unbalance - percent, maximum	Not applicable
Characteristic impedance - at 1 MHz, ohms	Not applicable
Watertightness (see Table IX of MIL-DTL-24640 for limits of water leakage)	No failure
Weight	No failure

Group B:

Cold bending	No failure
Drip	No failure
Cross link proof test (percent maximum)	
Jacket	50
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300

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Elongation - percent, minimum 160
Attenuation - at 3 MHz, dB/100 feet, maximum ..... Not applicable
Gas flame (1 hour) ..... No failure

Group C:
Physicals (aged)
Jacket
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged, minimum 60
Permanence of printing (conductor) - Method 1 only cycles, minimum 25
Permanence of printing (jacket) - cycles, minimum 125
Shield - conformance to material, construction and coverage (type FXOW only) ..... No failure
Heat distortion - percent of unaged, maximum 30
Cable sealant removability ..... No failure
Shrinkage ..... No failure
Mutual inductance - dB/1000 feet, maximum ..... Not applicable
Surface transfer impedance (type FXOW only)
Milliohms per meter, maximum 700
EMP response, dB, minimum 60
Pulse response time - ms/500 feet
Rise time, maximum ..... Not applicable
Decay time, maximum ..... Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable) ..... No failure
Acid gas equivalent - percent, maximum
Jacket 2
Fillers 2
Insulation 18
Halogen content - percent, maximum
Jacket 0.2
Fillers 0.2
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Smoke index, maximum
Jacket 25
Fillers 45
Insulation 10
Toxicity index, maximum

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Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure
Fungus Resistance	No failure
Abrasion Resistance	No failure
Gas Flame (3 hour)	No failure
Component Wire	No failure

Unit ordering length: 500 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2307-021)

Review Activities:
Army - AR, AV, CR, CR4
Navy - CG, EC
DLA - IS

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